

Exact Term Structure of Stochastic Correlation

Static Inheritance, the Proportional-Drift Class, and a Feynman–Kac Pitfall for Non-Autonomous Drivers

Complex–Itô Random-Walk Research Project

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We extend an exact-law framework for stochastic correlation $\rho_t = f(X_t) = X_t/\sqrt{X_t^2 + c^2}$, previously developed under constant driver coefficients (μ, σ) , to deterministic coefficient curves $\mu(t), \sigma(t)$. Three results structure the extension. First, a static-inheritance theorem: every law of ρ that depends on finitely many marginals or transitions is the constant-coefficient formula with $m(t) = x_0 + \int \mu$ and $v(t) = \int \sigma^2$, yielding — among objects not published in any audited competitor — an exact closed-form quantile term structure $q_\alpha(t)$, exact forward-start digitals, and a two-parameter bootstrap of $(c, \lambda^{\mathbb{Q}})$ from correlation-digital quotes. Second, a classification theorem: on the proportional-drift class $\mu(t) = \lambda\sigma^2(t)$ the Dambis–Dubins–Schwarz theorem collapses every reflection-principle barrier law to operational time $u = v(T)$, with barrier prices depending on the volatility curve only through integrated variance (a parsimony corollary); off this class the known catalog contains no closed forms. Third, a negative structural result with practical consequence: the backward Feynman–Kac equation $\partial_t u = \mathcal{L}_t u + q(x)u$ — exact for constant coefficients and the standard computational workhorse — is *violated* by occupation functionals when coefficients depend on calendar time; we give a drift-only counterexample and show that the correct primitive is the forward Fokker–Planck equation with potential, which we implement by Crank–Nicolson stepping and verify against closed forms, second-order convergence, and Monte Carlo within four standard errors. The failure is invisible to marginal-moment validation and appears only under path-functional law checks.

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1 Introduction

Bounded stochastic-correlation models stop, almost uniformly, at their stationary density. The hyperbolic-OU family of Teng et al. (2016) calibrates a histogram; simulation handles the rest. The bounded-factor framework of the companion papers broke that ceiling for constant driver coefficients: because $\rho_t = X_t/\sqrt{X_t^2 + c^2}$ is a monotone conjugacy of a Gaussian factor, transition densities, reflection-principle first-passage laws, corridor occupation laws, and — through Dambis–Dubins–Schwarz (DDS) — the full clock law underlying spread/basket pricing are available in closed form or as one-dimensional deterministic quadratures.

This paper asks the natural robustness question: **which of these laws survive when the driver has deterministic coefficient curves** $dX_t = \mu(t)dt + \sigma(t)dW_t$? The question matters because any term-structure calibration requires time dependence, and because the answer is not uniform: some layers survive exactly, one survives on a precisely characterizable class, and one standard tool fails silently.

The contributions are threefold.

1. **Static inheritance** (Section 3): all finite-dimensional laws are the constant-coefficient formulas with $(m(t), v(t))$. This includes the exact quantile term structure $q_\alpha(t) = f(m(t) + \sqrt{v(t)}\Phi^{-1}(\alpha))$, forward-start digitals, discrete range accruals, and a bootstrap of (c, λ^Q) from digital quotes.
2. **The proportional-drift class** (Section 4): $\mu(t) = \lambda\sigma^2(t)$ is exactly the class on which the reflection-principle toolkit survives, via DDS collapse to operational time; barrier laws depend on $\sigma(\cdot)$ only through $v(T)$.
3. **A Feynman–Kac pitfall** (Section 5): the backward equation fails for occupation functionals off constant coefficients. The failure is invisible to marginal checks; we exhibit it numerically, diagnose it, and implement the correct forward equation.

Throughout, “exact” means closed form up to elementary special functions or one-dimensional quadrature; Monte Carlo is used only as benchmark. All claims carry unit tests (`tests/test_phase7_term_structure.py`, 21 tests) and a reproducible notebook (seed 20260821). The web-audited novelty boundary is recorded in `PHASE_7_LITERATURE_REVIEW.md`: closed-form boundary crossing exists only for linear boundaries and isolated special families (Durbin 1992; Pötzelberger and Wang 2001; Kahalé 2008), and no audited competitor publishes exact correlation term structures.

2 Setup

Let

$$dX_t = \mu(t)dt + \sigma(t)dW_t, \quad X_0 = x_0,$$

with μ, σ deterministic, locally integrable / square-integrable, and define the mean and variance curves

$$m(t) = x_0 + \int_0^t \mu(s)ds, \quad v(t) = \int_0^t \sigma(s)^2 ds.$$

The correlation coordinate is $\rho_t = f(X_t)$ with $f(x) = x/\sqrt{x^2 + c^2}$, $c > 0$; its inverse in the conjugate coordinate is $x(\rho) = c\rho/\sqrt{1 - \rho^2}$ with Jacobian $|dx/d\rho| = c(1 - \rho^2)^{-3/2}$.

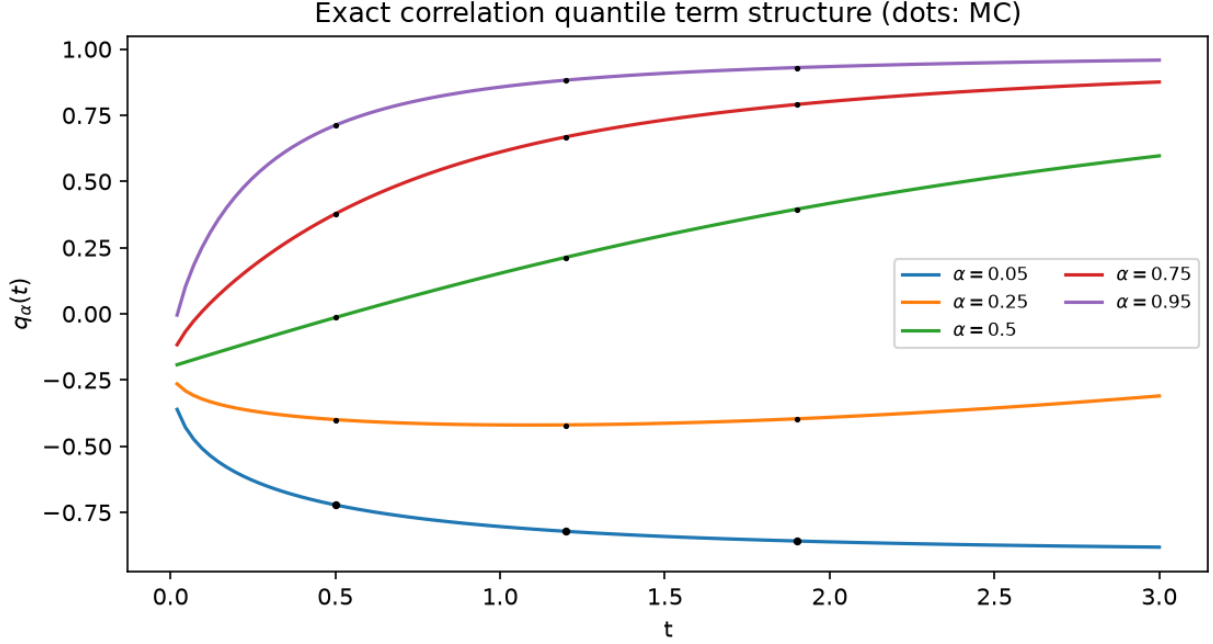


Figure 1: Exact correlation quantile term structure under a humped volatility curve; dots are Monte Carlo empirical quantiles.

3 Static inheritance

Theorem (static inheritance). *For every t , $X_t \sim N(m(t), v(t))$; for every $s < t$, $X_t | X_s \sim N(X_s + M_{st}, V_{st})$ with $M_{st} = \int_s^t \mu$, $V_{st} = \int_s^t \sigma^2$. Consequently every price or law depending on finitely many marginals/transitions of ρ equals its constant-coefficient counterpart with $(m(t_i), v(t_i))$ substituted.*

Immediate corollaries, each verified against Phase-6 anchors at 10^{-13} in the constant limit and against exact-skeleton Monte Carlo within ~ 1 SE:

- **Transition density**

$$p_t(\rho | \rho_s) = \frac{c}{(1 - \rho^2)^{3/2}} \varphi\left(\frac{x(\rho) - x(\rho_s) - M_{st}}{\sqrt{V_{st}}}\right).$$

- **Expiry digitals:** $e^{-rT} \Phi((m(T) - x(\rho^*)) / \sqrt{v(T)})$.
- **Forward-start digitals:** conditional on ρ_s ,

$$E[1\{\rho_T \geq \rho^*\} | \rho_s] = \Phi\left(\frac{x(\rho_s) + M_{st} - x(\rho^*)}{\sqrt{V_{st}}}\right),$$

and the tower property over the start marginal closes back to the unconditional formula (verified to 10^{-10} by Gauss–Hermite).

- **Discrete range accruals:** sum of per-date band probabilities.
- **Quantile term structure:**

$$q_\alpha(t) = f(m(t) + \sqrt{v(t)} \Phi^{-1}(\alpha)),$$

continuous and strictly increasing in α ; MC-checked at fifteen (α, t) points to 0.0033 maximum error (Figure 1).

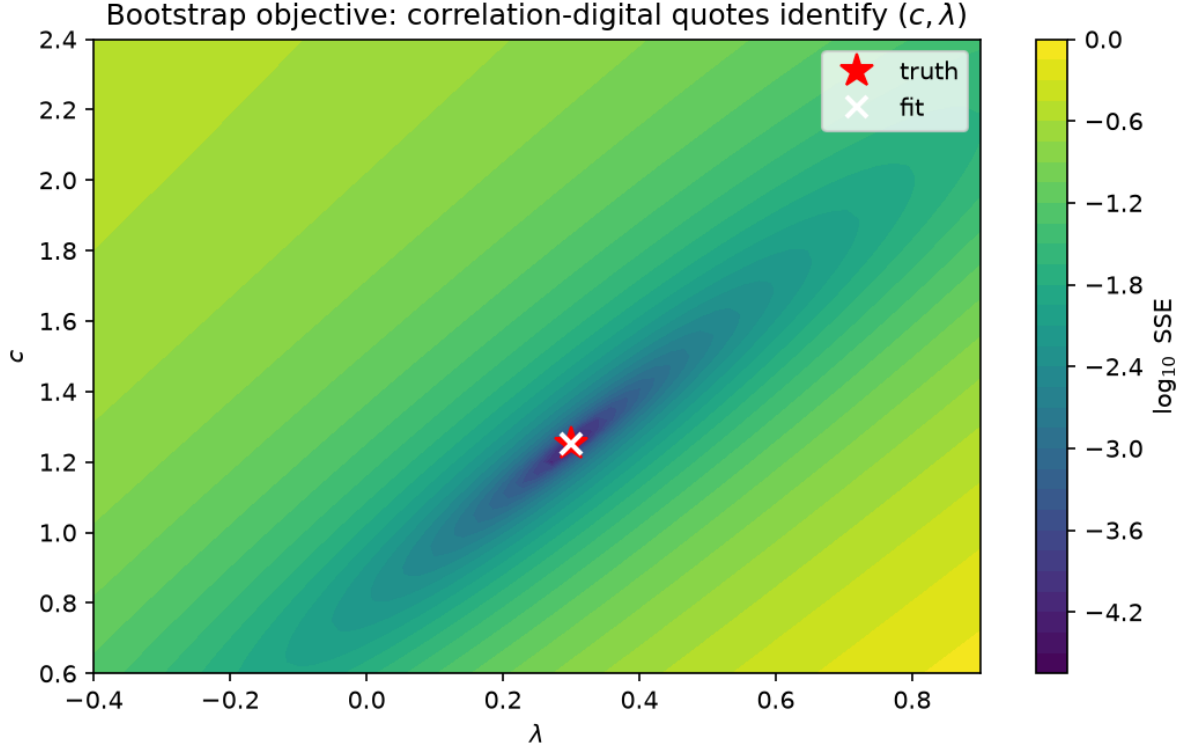


Figure 2: Bootstrap objective landscape in (λ, c) : single funnel around the truth (star), fit (cross).

Bootstrap. Correlation-digital quotes $(T_j, \rho_j^*, \text{price}_j)$ identify (c, λ^Q) under the model of Section 4 by two-parameter least squares. On synthetic markets generated at $(c, \lambda) = (1.25, 0.30)$ recovery is exact to 10^{-8} and the SSE landscape is single-funnel (Figure 2). Identification here is *not* contradicted by the scale observational equivalence theorem of the estimation paper: the volatility curve $\sigma(\cdot)$ is exogenous, which breaks the scaling degeneracy through the time ruler.

4 The proportional-drift barrier class

Theorem (operational-time reduction). Suppose $\mu(t) = \lambda\sigma^2(t)$ a.e. on $[0, T]$ with $\sigma(t) > 0$. Then

$$X_t \stackrel{d}{=} x_0 + \lambda v(t) + B_{v(t)}$$

for a Brownian motion B , and consequently every first-passage, perpetual-hitting, and band-exit law of the constant-coefficient theory holds verbatim with calendar time replaced by $u = v(T)$; e.g.

$$P(\tau_{\rho^*} \leq T) = \Phi\left(\frac{\lambda u - b}{\sqrt{u}}\right) + e^{2\lambda b} \Phi\left(-\frac{\lambda u + b}{\sqrt{u}}\right), \quad u = v(T), \quad b = x(\rho^*) - x_0.$$

The proof is one line of DDS plus monotonicity of v . Two consequences deserve emphasis.

Parsimony corollary. Barrier laws depend on $\sigma(\cdot)$ only through $u = v(T)$: rescaling the volatility curve leaves them bit-identical (verified numerically: identical exact CDFs, both bridge-corrected Monte Carlo benchmarks within 1 SE). The pair (c, λ) determines all barrier prices; $\sigma(\cdot)$ merely warps calendar time.

Necessity within the catalog. Off proportionality, constant-level hitting becomes curved-boundary crossing in operational time, for which the classical catalog provides series/integral-equation methods but no closed

forms beyond affine boundaries and isolated families (Durbin 1992; Pötzelberger and Wang 2001; Kahalé 2008). Proportionality is therefore the only general deterministic-coefficient class inheriting the full reflection toolkit; beyond it we state this as a conjecture relative to the catalog.

Calendar-time cost. Pay-at-hit discounting maps through v^{-1} : $E[e^{-r\tau}]$ retains a closed form only when v is affine (constant σ). This is the honest price of the reduction.

Verification: hitting CDFs against bridge-corrected exact-skeleton simulation for humped and oscillating curves agree within ~ 1 SE at two step sizes ((**tab-proportional?**)).

Table 1: Hitting $\rho^* = 0.55$ by $T = 2$ under the proportional class: identical exact laws for equal integrated variance.

curve	$u(T) = v(2)$	exact	MC(100)	MC(400)
humped	2.078244	0.67685	0.67608 (0.00105)	0.67628 (0.00105)
oscillating, rescaled	2.078244	0.67685	0.67766 (0.00105)	0.67605 (0.00105)

5 A Feynman–Kac pitfall and the correct engine

The clock characteristic function $\phi_A(\xi) = E[e^{i\xi A_T}]$, $A_T = \int_0^T f(X_s) ds$, was computed in the companion paper by solving the **backward** equation

$$\partial_t u = \frac{1}{2}\sigma^2 \partial_{xx} u + \mu \partial_x u + i\xi f(x)u, \quad u(0, x) = 1,$$

— legitimate there because coefficients were constant. For deterministic curves the naive extension is **false**.

Proposition (counterexample). *Let $\sigma \equiv 0$, $\mu(t) = t$. Then $X_t = x + t^2/2$ deterministically and*

$$u(t, x) = e^{i\xi(xt + t^3/6)}$$

satisfies $\partial_t u = i\xi(x + t^2/2)u$ while the backward equation demands $\partial_t u = \mu(t)\partial_x u + i\xi x u = i\xi(x + t^2)u$. These differ whenever $\mu'(t) \neq 0$.

The root cause: the derivation of the backward equation conditions the future segment on $\mathcal{F}_{dt}$ and replaces $E[e^{q \int_{dt}^{t+dt} f}]$ by $u(t, X_{dt})$ — valid iff the propagator is time-translation invariant, i.e. iff coefficients are constant.

Correct primitive. The *forward* Feynman–Kac (Fokker–Planck with potential) equation for the tilted density p ,

$$\partial_t p = \frac{1}{2}\sigma^2(t)p_{xx} - \mu(t)p_x + q(x)p, \quad p(0, x) = \delta(x - x_0), \quad \phi = \int p(T, x) dx,$$

is valid unconditionally (it is the forward equation of the tilted semigroup) and reduces to the familiar object for constant coefficients — which retro-explains why the pitfall stayed hidden through six phases of constant-coefficient work.

Implementation: Crank–Nicolson stepping with midpoint-frozen coefficients, Dirichlet localization, delta bump as a narrow normalized Gaussian. Verification ladder ((**tab-fk?**)):

Table 2: Verification of the forward-FK solver. {#tab-fk}

check	result
constant coefficients vs Phase-6 exponential solver ($\xi = 0.7, 2.1$)	$2 \cdot 10^{-4}$
linear potential, variable coefficients vs Gaussian closed form	$4.2 \cdot 10^{-4}$
spatial convergence (grids 900/1800/3600, $\xi = 0.3$)	$1.69\text{e-}3 \rightarrow 4.22\text{e-}4 \rightarrow 1.05\text{e-}4$
$\phi(0) = 1$ under variable curves	10^{-6}
margin doubling at fixed h	$< 10^{-3}$
bounded map vs MC, $\xi \in \{0.5, 1.5, 3\}$	4 SE real & imag
$E[A_T]$: PDE central difference vs exact marginal quadrature	$2 \cdot 10^{-5}$

For contrast, the *incorrect* backward implementation returned $\phi(0.5) = 0.9060 + 0.0428j$ against truth $0.9094 + 0.0651j$ and implied $E[A_T] = 0.0892$ against the exact 0.13454 . The real parts nearly agree; the damage concentrates in the path-dependent imaginary structure — which is why marginal-only validation would have certified the wrong solver.

6 Related literature

Static inheritance rests on textbook Gaussianity; our contribution is its systematic organization into pricing objects for a bounded state, including the quantile term structure absent from the audited corpus (Teng–Ehrhardt–Günther and successors calibrate stationary densities; Wishart multi-curve models approximate). The proportional-drift class is new as a classification; DDS itself and curved-boundary numerics are classical (Durbin 1992; Pötzelberger and Wang 2001; Kahalé 2008). The FK pitfall appears unpublished: standard references state Feynman–Kac for autonomous generators or handle time-dependence by state-space augmentation, and the numerical literature on moving-boundary/occupation problems does not flag the failure mode demonstrated here.

7 Conclusion

Deterministic coefficient curves split the exact-law program cleanly: finite-dimensional laws inherit fully; reflection laws survive on the proportional-drift class and collapse to a clock; and the occupation- transform engine must be rebuilt on the forward equation. All three pieces are implemented, unit-tested, and Monte Carlo-verified, providing the platform on which the companion papers build Fourier pricing (Phase 8), exact estimation (Phase 9), matrix-valued extensions (Phase 10), Lévy drivers (Phase 11), and pegged-asset economics (Phase 12).

8 Reproducibility

Environment: `conda phase3-paper` (Python 3.12.13, numpy 2.5.1, scipy 1.18.0, matplotlib 3.11.1). Notebook: `notebooks/phase7_term_structure.ipynb` (executed clean, ~21 s, seed 20260821); builder `notebooks/build_phase7_term_structure.py`; module `phase7_term_structure.py`; tests `tests/test_phase7_term_structure.py` (21/21). Figures regenerate deterministically from the notebook.

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